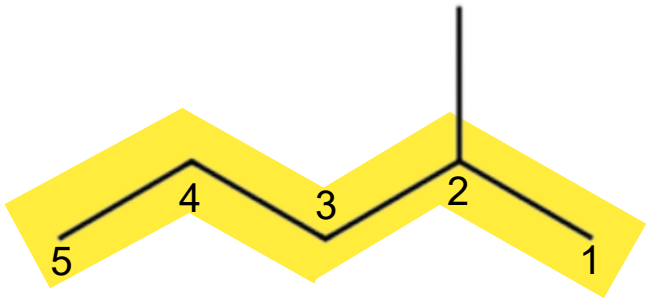
A01 — STRUCTURE → NAME

Level: FOUNDATION **MOLECULAR FORMULA:** C_6H_{14}

NUMBERED PARENT



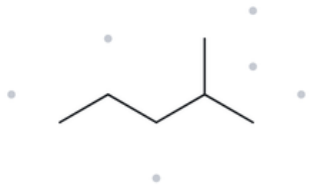
NAMING TRAIL

PARENT: 5C so pentane
NUMBERING: 2 since 2 is much closer in substituent than 4
SUBSTITUENT: methyl

MY NAME

2-methylpentane

THE ALKINE CHECK

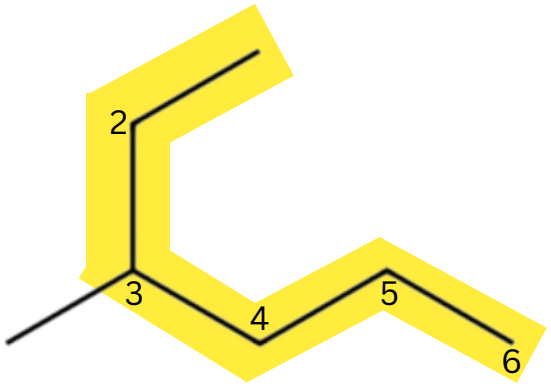


2-methylpentane

A02 — STRUCTURE → NAME

Level: FOUNDATION **MOLECULAR FORMULA:** C_7H_{16}

NUMBERED PARENT



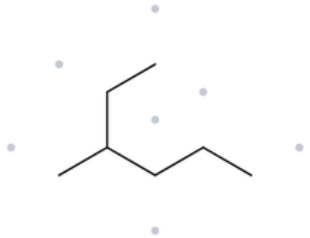
NAMING TRAIL

PARENT: 6C so HEXANE
NUMBERING: 3 since 3 is much closer in substituent than 4
SUBSTITUENT: methyl

MY NAME

3-methylhexane

THE ALKINE CHECK

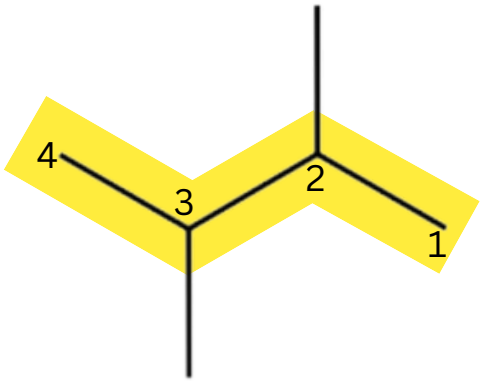


3-methylhexane

A03 — STRUCTURE → NAME

Level: FOUNDATION **MOLECULAR FORMULA:** C_6H_{14}

NUMBERED PARENT



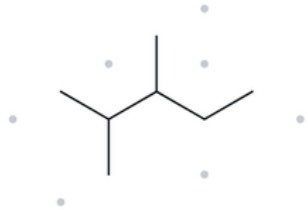
NAMING TRAIL

PARENT: 4C so PENTANE
NUMBERING: it is from right to left
SUBSTITUENT: methyl

MY NAME

2,3-dimethylpentane

THE ALKINE CHECK

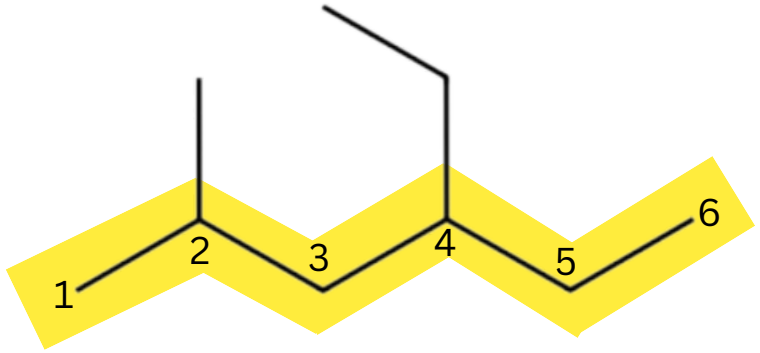


2,3-dimethylpentane

A04 — STRUCTURE → NAME

Level: DEVELOPING **MOLECULAR FORMULA:** C_9H_{20}

NUMBERED PARENT



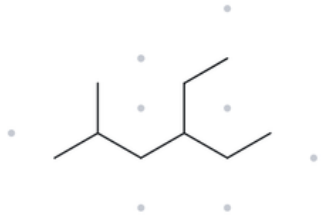
NAMING TRAIL

PARENT: 6C so HEXANE
NUMBERING: it is from left to right
SUBSTITUENT: methyl is on C2 and ethyl on C4

MY NAME

4-ethyl-2-methylhexane

THE ALKINE CHECK

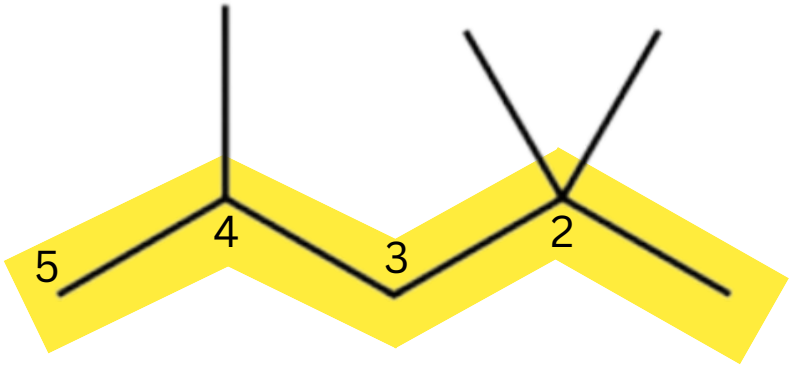


4-ethyl-2-methylhexane

A05 — STRUCTURE → NAME

Level: DEVELOPING **MOLECULAR FORMULA:** C_8H_{18}

NUMBERED PARENT



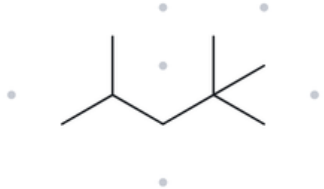
NAMING TRAIL

PARENT: 5C so pentane
NUMBERING: it is from right to left
SUBSTITUENT: there are 2 methyl on C2 and 1 on C4

MY NAME

2,2,4-trimethylpentane

THE ALKINE CHECK

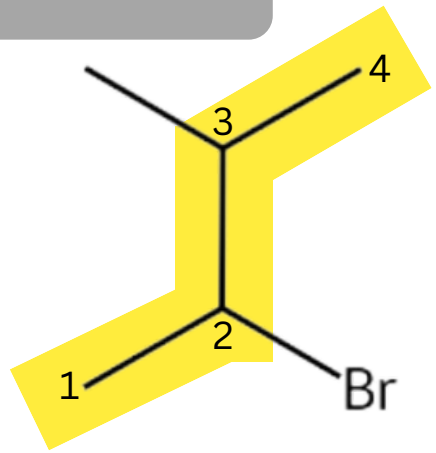


2,2,4-trimethylpentane

A06 — STRUCTURE → NAME

Level: DEVELOPING **MOLECULAR FORMULA: $C_5H_{11}Br$**

NUMBERED PARENT



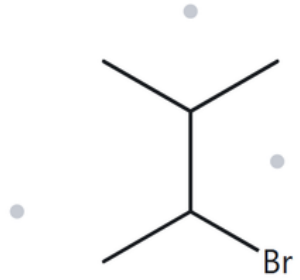
NAMING TRAIL

PARENT: 4C so butane
NUMBERING: both directions give the same so doing it alphabetically Br gets lowest
SUBSTITUENT: methyl at C3 and Br at C2

MY NAME

2-bromo-3-methylbutane

THE ALKINE CHECK

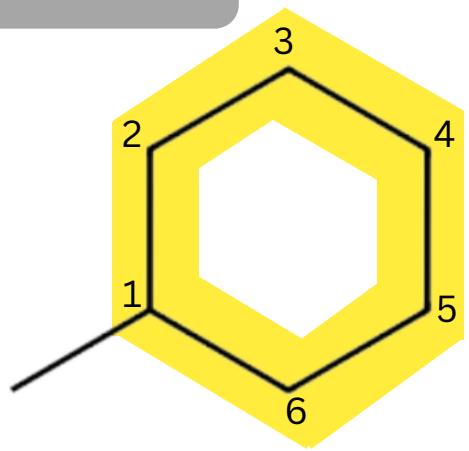


2-bromo-3-methylbutane

A07 — STRUCTURE → NAME

Level: DEVELOPING **MOLECULAR FORMULA:** C_7H_{14}

NUMBERED PARENT



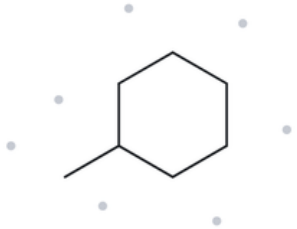
NAMING TRAIL

PARENT: 6C so cyclohexane
NUMBERING: not required since ring
SUBSTITUENT: methyl

MY NAME

methylcyclohexane

THE ALKINE CHECK

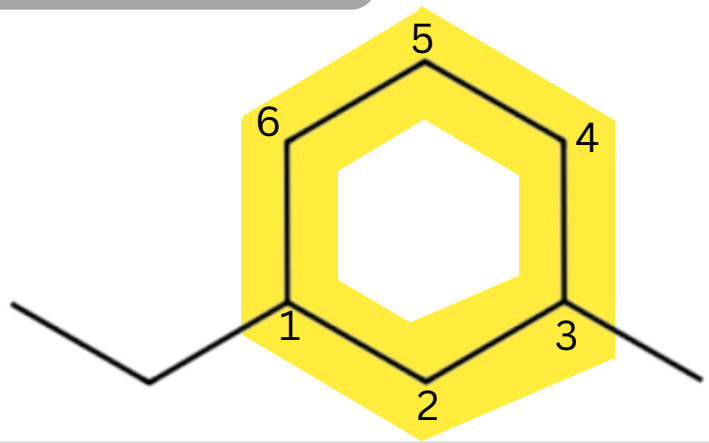


methylcyclohexane

A08 — STRUCTURE → NAME

Level: CHALLENGE **MOLECULAR FORMULA:** C_9H_{18}

NUMBERED PARENT



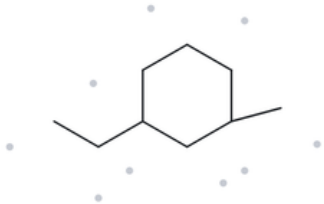
NAMING TRAIL

PARENT: 6C so cyclohexane
NUMBERING: they are opposite so alphabetically ethyl is first
SUBSTITUENT: ethyl is at C1 and methyl is at C4

MY NAME

1-ethyl-3-methylcyclohexane

THE ALKINE CHECK

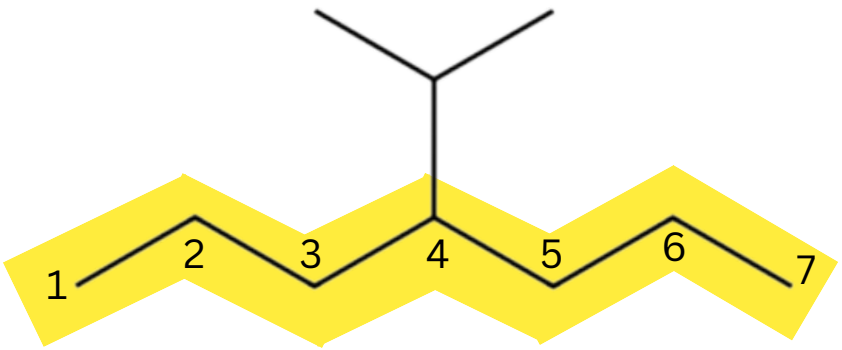


1-ethyl-3-methylcyclohexane

A09 — STRUCTURE → NAME

Level: CHALLENGE **MOLECULAR FORMULA:** $C_{10}H_{22}$

NUMBERED PARENT



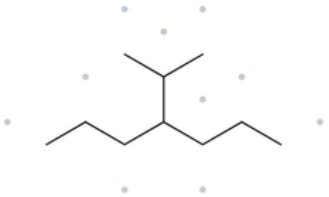
NAMING TRAIL

PARENT: 7C so heptane
NUMBERING: either direction
SUBSTITUENT: methyl is at C2 and isoprophyl is at C4

MY NAME

2-methyl-3-propylhexane

THE ALKINE CHECK

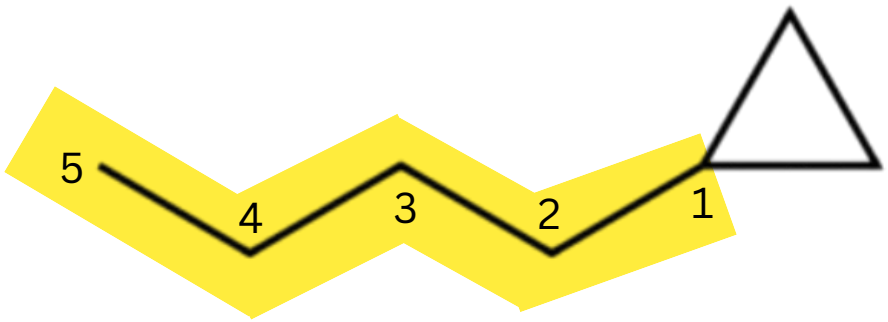


2-methyl-3-propylhexane

A10 — STRUCTURE → NAME

Level: CHALLENGE **MOLECULAR FORMULA:** C_8H_{16}

NUMBERED PARENT



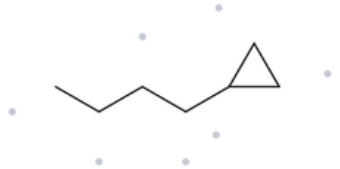
NAMING TRAIL

PARENT: 3C so cyclopropane
NUMBERING: not required a locant
SUBSTITUENT: pentyl group

MY NAME

butylcyclopropane

THE ALKINE CHECK



butylcyclopropane

CARD B01 — NAME → STRUCTURE

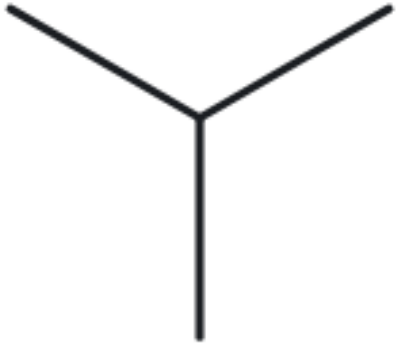
Level: FOUNDATION **Given name:** 2-methylpropane

NAME DECODED

PARENT: propane means a 3 carbon chain
SUBSTITUENT: methyl on C1
EXPECTED FORMULA: 4 carbons since 3 from parent and 1 from chain and 10H since 3 carbons has 3 each and 1 carbon has 1H=10

C_4H_{10}

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: C_4H_{10}
CHECKING: C_NH_{2N+2} = $C_4H_{2(4)+2}$ = C_4H_{10}

BACK-CHECK

RETURNED BY THE ALKANE: 2-methylpropane

CARD B02 — NAME → STRUCTURE

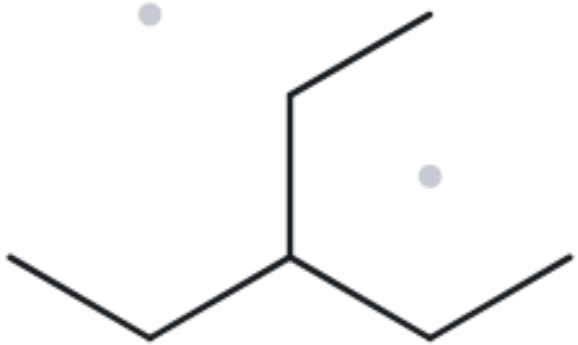
Level: FOUNDATION **Given name:** 3-ethylpentane

NAME DECODED

PARENT: pentane means a 5 carbon chain
SUBSTITUENT: ethyl on C3
EXPECTED FORMULA: 7 carbons since 7 chains and 16H since 2(7)+2=16

C_7H_{16}

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: C_7H_{16}
CHECKING: $C_NH_{2N+2} = C_7H_{2(7)+2} =$

C_7H_{16}

BACK-CHECK

RETURNED BY THE ALKANE: 3-ethylpentane

CARD B03 — NAME → STRUCTURE

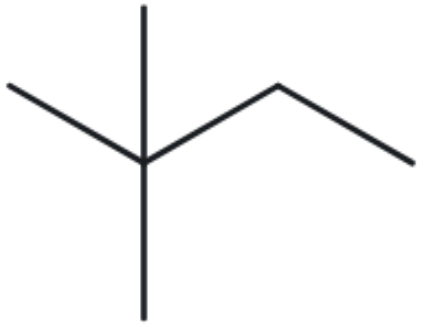
Level: FOUNDATION **Given name:** 2,2-dimethylbutane

NAME DECODED

PARENT: butane means a 4 carbon chain
SUBSTITUENT: methyl on C2 and di so it means 2 methyl branch
EXPECTED FORMULA: 6 carbons since 6 chains and 14H since 2(6)+2=14

C_6H_{14}

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: C_6H_{14}
CHECKING: C_NH_{2N+2} = $C_6H_{2(6)+2}$ = C_6H_{14}

BACK-CHECK

RETURNED BY THE ALKANE: 2,2-dimethylbutane

CARD B04 — NAME → STRUCTURE

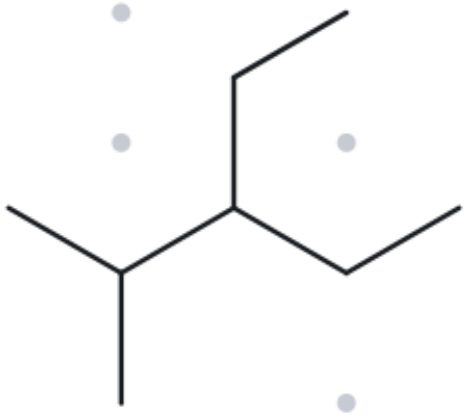
Level: DEVELOPING **Given name:** 3-ethyl-2-methylpentane

NAME DECODED

PARENT: pentane means a 5 carbon chain
SUBSTITUENT: methyl on C2 and ethyl on C3
EXPECTED FORMULA: 8 carbons since 8 chains and 14H since 2(8)+2=18

C_8H_{18}

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: C_8H_{18}
CHECKING: $C_NH_{2N+2} = C_8H_{2(8)+2} =$

C_8H_{18}

BACK-CHECK

RETURNED BY THE ALKANE: 2,2-dimethylbutane

CARD B05 — NAME → STRUCTURE

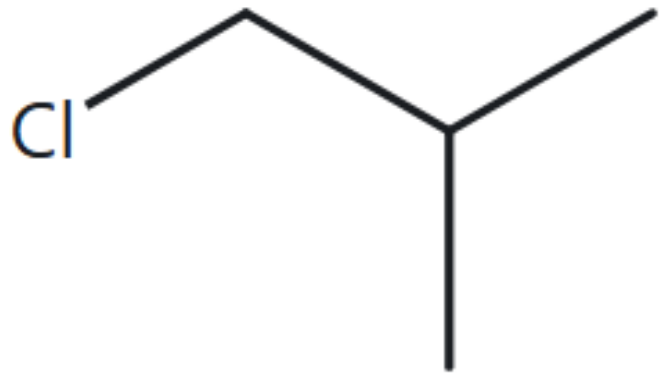
Level: DEVELOPING **Given name:** 1-chloro-2-methylpropane

NAME DECODED

PARENT: propane means a 3 carbon chain
SUBSTITUENT: methyl on C2 and chlorine on C1
EXPECTED FORMULA: 4 carbons since 4 chains Cl not included and 10H since 2(4)+2=10

C_4H_{10}

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: C_4H_{10}
CHECKING: C_NH_{2N+2} = $C_4H_{2(4)+2}$ = C_4H_{10}

BACK-CHECK

RETURNED BY THE ALKANE: 1-chloro-2-methylpropane

CARD B06 — NAME → STRUCTURE

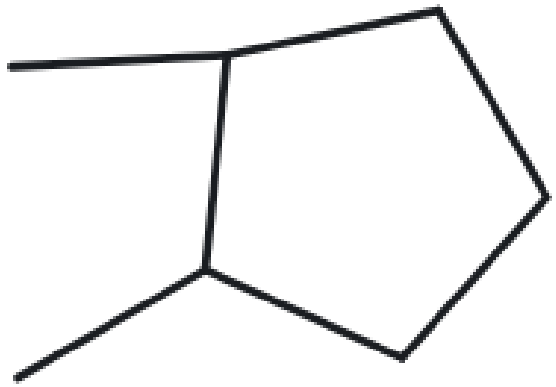
Level: DEVELOPING **Given name:** 1,2-dimethylcyclopentane

NAME DECODED

PARENT: cyclopentane means a ring containing 5 carbons
SUBSTITUENT: methyl on C2 and C1
EXPECTED FORMULA: 7 carbons since 5 on the ring and 2 on methyl and 14H since 2(7)=14

C_7H_{14}

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: C_7H_{14}
CHECKING: C_NH_{2N} = $C_7H_{2(7)}$ =

C_7H_{14}

BACK-CHECK

RETURNED BY THE ALKANE: 1,2-dimethylcyclopentane

CARD B07 — NAME → STRUCTURE

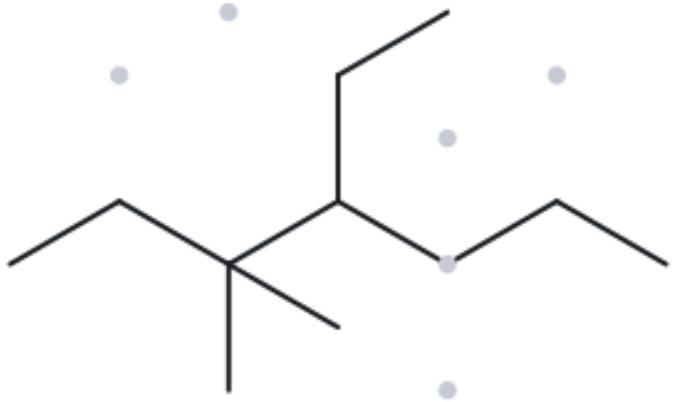
Level: DEVELOPING **Given name:** 4-ethyl-3,3-dimethylheptane

NAME DECODED

PARENT: heptane means a 7 carbon chain
SUBSTITUENT: 2 methyl on C3 so di- and ethyl on C4
EXPECTED FORMULA: 11 carbons since 11 chains and 24H since 2(11)+2=24

$C_{11}H_{24}$

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: $C_{11}H_{24}$
CHECKING: $C_NH_{2N+2} = C_{11}H_{2(11)+2} =$

$C_{11}H_{24}$

BACK-CHECK

RETURNED BY THE ALKANE: 4-ethyl-3,3-dimethylheptane

CARD B08 — NAME → STRUCTURE

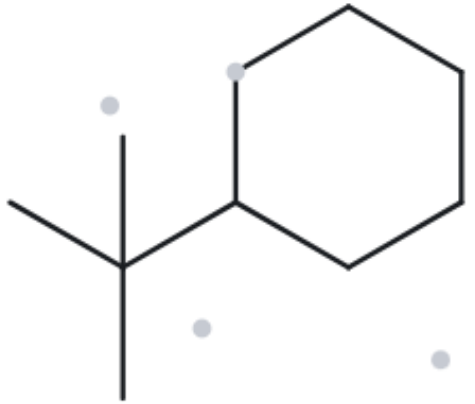
Level: CHALLENGE **Given name:** tert-butylcyclohexane

NAME DECODED

PARENT: cyclohexane means a ring containing 6 carbons
SUBSTITUENT: turt-butyl
EXPECTED FORMULA: 10 carbons since 6 on the ring and 4 on turt-butyl and 20H since 2(10)=20

$C_{10}H_{20}$

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: $C_{10}H_{20}$
CHECKING: C_NH_{2N} = $C_{10}H_{2(10)}$ =

$C_{10}H_{20}$

BACK-CHECK

RETURNED BY THE ALKANE: (1,1-dimethylethyl)cyclohexane or tert-butylcyclohexane

CARD B09 — NAME → STRUCTURE

Level: CHALLENGE **Given name:** 3-ethyl-1,1-dimethylcyclohexane

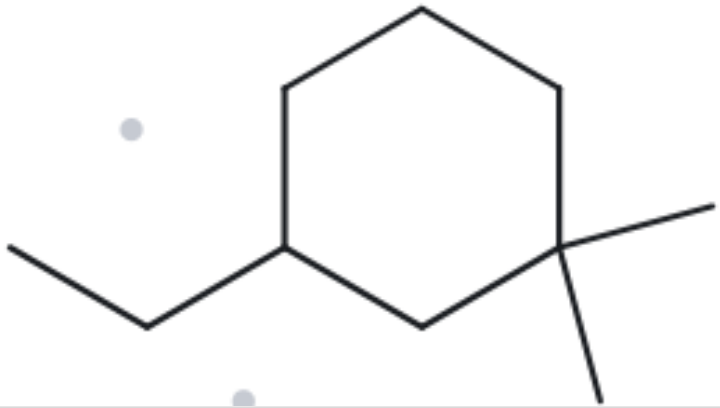
NAME DECODED

PARENT: cyclohexane means a ring containing 6 carbons

SUBSTITUENT: 2 methyl in C1 so di- and ethyl in C3

EXPECTED FORMULA: 10 carbons since 6 on the ring and 4 from both metyl and ethyl and 20H since 2(10)=20 $C_{10}H_{20}$

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: $C_{10}H_{20}$

CHECKING: C_NH_{2N} = $C_{10}H_{2(10)}$ = $C_{10}H_{20}$

BACK-CHECK

RETURNED BY THE ALKANE: 3-ethyl-1,1-dimethylcyclohexane

CARD B10 — NAME → STRUCTURE

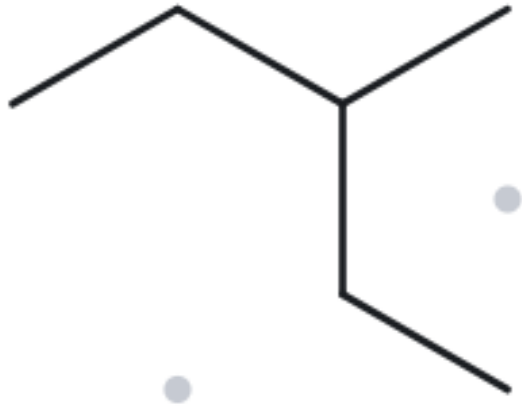
Level: CHALLENGE **Given name:** 2-ethylbutane

NAME DECODED

PARENT: pentane means a 5 carbon chain
SUBSTITUENT: methyl on C3
EXPECTED FORMULA: 6 carbons since 6 chains and 14H since 2(6)+2=14

C_6H_{14}

BOND-LINE STRUCTURE



FORMULA CHECK

COUNTED FROM DRAWING: C_6H_{14}
CHECKING: $C_NH_{2N+2} = C_6H_{2(6)+2} =$

C_6H_{14}

BACK-CHECK

RETURNED BY THE ALKANE: 3-methylpentane